

Newman-Penrose Formalism Calculations

Stephani et al., Equation (20.3)

Metric

$$ds^2 = -e^{2U} dt \otimes dt - e^{2U} A dt \otimes d\varphi - e^{2U} A d\varphi \otimes dt + \left(-\frac{e^{4U} A^2 - \rho^2}{e^{2U}} \right) d\varphi \otimes d\varphi + (e^{-2U+2k}) d\rho \otimes d\rho + (e^{-2U+2k}) dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, \varphi, \rho, z)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2} e^U \right) dt + \left(\frac{1}{2} \sqrt{2} e^U A + \frac{\sqrt{2} \rho}{2 e^U} \right) d\varphi$$

$$n = \left(\frac{1}{2} \sqrt{2} e^U \right) dt + \left(\frac{1}{2} \sqrt{2} e^U A - \frac{\sqrt{2} \rho}{2 e^U} \right) d\varphi$$

$$m = \left(\frac{\sqrt{2} e^k}{2 e^U} \right) d\rho + \left(\frac{i \sqrt{2} e^k}{2 e^U} \right) dz$$

$$\bar{m} = \left(\frac{\sqrt{2} e^k}{2 e^U} \right) d\rho + \left(-\frac{i \sqrt{2} e^k}{2 e^U} \right) dz$$

Contravariant components

$$l = \left(-\frac{\sqrt{2} e^{2U} A}{2 e^U \rho} - \frac{\sqrt{2}}{2 e^U} \right) dt + \left(\frac{\sqrt{2} e^U}{2 \rho} \right) d\varphi$$

$$n = \left(\frac{\sqrt{2}e^{2U}A}{2e^U\rho} - \frac{\sqrt{2}}{2e^U} \right) dt + \left(-\frac{\sqrt{2}e^U}{2\rho} \right) d\varphi$$

$$m = \left(\frac{\sqrt{2}e^U}{2e^k} \right) d\rho + \left(\frac{i\sqrt{2}e^U}{2e^k} \right) dz$$

$$\bar{m} = \left(\frac{\sqrt{2}e^U}{2e^k} \right) d\rho + \left(-\frac{i\sqrt{2}e^U}{2e^k} \right) dz$$

Directional Derivatives

$$D = -\frac{\sqrt{2}e^{2U}A}{2e^U\rho} - \frac{\sqrt{2}}{2e^U} \partial_t + \frac{\sqrt{2}e^U}{2\rho} \partial_\varphi$$

$$\Delta = \frac{\sqrt{2}e^{2U}A}{2e^U\rho} - \frac{\sqrt{2}}{2e^U} \partial_t - \frac{\sqrt{2}e^U}{2\rho} \partial_\varphi$$

$$\delta = \frac{\sqrt{2}e^U}{2e^k} \partial_\rho + \frac{i\sqrt{2}e^U}{2e^k} \partial_z$$

$$\bar{\delta} = \frac{\sqrt{2}e^U}{2e^k} \partial_\rho - \frac{i\sqrt{2}e^U}{2e^k} \partial_z$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = -\frac{\sqrt{2}e^U \log(e) \frac{\partial}{\partial \rho} U}{2e^k} - \frac{i\sqrt{2}e^U \log(e) \frac{\partial}{\partial z} U}{2e^k} + \frac{\sqrt{2}e^{3U} \frac{\partial}{\partial \rho} A}{4e^k \rho} + \frac{i\sqrt{2}e^{3U} \frac{\partial}{\partial z} A}{4e^k \rho} + \frac{\sqrt{2}e^U}{4e^k \rho}$$

$$\mu = 0$$

$$\lambda = 0$$

$$\nu = \frac{\sqrt{2}e^U \log(e) \frac{\partial}{\partial \rho} U}{2e^k} - \frac{i\sqrt{2}e^U \log(e) \frac{\partial}{\partial z} U}{2e^k} + \frac{\sqrt{2}e^{3U} \frac{\partial}{\partial \rho} A}{4e^k \rho} - \frac{i\sqrt{2}e^{3U} \frac{\partial}{\partial z} A}{4e^k \rho} - \frac{\sqrt{2}e^U}{4e^k \rho}$$

$$\alpha = \frac{\sqrt{2}e^U \log(e) \frac{\partial}{\partial \rho} U}{4e^k} - \frac{i\sqrt{2}e^U \log(e) \frac{\partial}{\partial z} U}{4e^k} - \frac{\sqrt{2}e^U \log(e) \frac{\partial}{\partial \rho} k}{4e^k} + \frac{i\sqrt{2}e^U \log(e) \frac{\partial}{\partial z} k}{4e^k} + \frac{\sqrt{2}e^{3U} \frac{\partial}{\partial \rho} A}{8e^k \rho} - \frac{i\sqrt{2}e^{3U} \frac{\partial}{\partial z} A}{8e^k \rho}$$

$$\beta = -\frac{\sqrt{2}e^U \log(e) \frac{\partial}{\partial \rho} U}{4e^k} - \frac{i\sqrt{2}e^U \log(e) \frac{\partial}{\partial z} U}{4e^k} + \frac{\sqrt{2}e^U \log(e) \frac{\partial}{\partial \rho} k}{4e^k} + \frac{i\sqrt{2}e^U \log(e) \frac{\partial}{\partial z} k}{4e^k} + \frac{\sqrt{2}e^{3U} \frac{\partial}{\partial \rho} A}{8e^k \rho} + \frac{i\sqrt{2}e^{3U} \frac{\partial}{\partial z} A}{8e^k \rho}$$

$$\tau = -\frac{\sqrt{2}e^U}{4e^k \rho}$$

$$\pi = \frac{\sqrt{2}e^U}{4e^k \rho}$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{e^{4U} \log(e) \frac{\partial}{\partial \rho} A \frac{\partial}{\partial \rho} U}{e^{2k} \rho} + \frac{e^{2U} \log(e) \frac{\partial^2}{(\partial \rho)^2} U}{2e^{2k}} - \frac{e^{4U} \log(e) \frac{\partial}{\partial z} A \frac{\partial}{\partial z} U}{e^{2k} \rho} + \frac{e^{2U} \log(e) \frac{\partial^2}{(\partial z)^2} U}{2e^{2k}} + \frac{e^{2U} \log(e) \frac{\partial}{\partial \rho} U}{2e^{2k} \rho} + \frac{e^{6U} \frac{\partial}{\partial \rho} A^2}{4e^{2k} \rho^2} - \frac{e^{4U} \frac{\partial^2}{(\partial \rho)^2} A}{4e^{2k} \rho} + \frac{e^{6U} \frac{\partial}{\partial z} A^2}{4e^{2k} \rho^2} - \frac{e^{4U} \frac{\partial^2}{(\partial z)^2} A}{4e^{2k} \rho} + \frac{e^{4U} \frac{\partial}{\partial \rho} A}{4e^{2k} \rho^2}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = -\frac{e^{2U} \log(e)^2 \frac{\partial}{\partial \rho} U^2}{2e^{2k}} - \frac{i e^{2U} \log(e)^2 \frac{\partial}{\partial \rho} U \frac{\partial}{\partial z} U}{e^{2k}} + \frac{e^{2U} \log(e)^2 \frac{\partial}{\partial z} U^2}{2e^{2k}} + \frac{e^{2U} \log(e) \frac{\partial}{\partial \rho} k}{2e^{2k} \rho} + \frac{i e^{2U} \log(e) \frac{\partial}{\partial z} k}{2e^{2k} \rho} + \frac{e^{6U} \frac{\partial}{\partial \rho} A^2}{8e^{2k} \rho^2} + \frac{i e^{6U} \frac{\partial}{\partial \rho} A \frac{\partial}{\partial z} A}{4e^{2k} \rho^2} - \frac{e^{6U} \frac{\partial}{\partial z} A^2}{8e^{2k} \rho^2}$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{e^{2U} \log(e)^2 \frac{\partial}{\partial \rho} U^2}{4e^{2k}} - \frac{e^{2U} \log(e)^2 \frac{\partial}{\partial z} U^2}{4e^{2k}} + \frac{e^{2U} \log(e) \frac{\partial^2}{(\partial \rho)^2} U}{4e^{2k}} + \frac{e^{2U} \log(e) \frac{\partial^2}{(\partial z)^2} U}{4e^{2k}} - \frac{e^{2U} \log(e) \frac{\partial^2}{(\partial \rho)^2} k}{4e^{2k}} - \frac{e^{2U} \log(e) \frac{\partial^2}{(\partial z)^2} k}{4e^{2k}} + \frac{e^{2U} \log(e) \frac{\partial}{\partial \rho} U}{4e^{2k} \rho} + \frac{e^{6U} \frac{\partial}{\partial \rho} A^2}{16e^{2k} \rho^2} + \frac{e^{6U} \frac{\partial}{\partial z} A^2}{16e^{2k} \rho^2}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = -\frac{e^{2U} \log(e)^2 \frac{\partial}{\partial \rho} U^2}{2e^{2k}} + \frac{i e^{2U} \log(e)^2 \frac{\partial}{\partial \rho} U \frac{\partial}{\partial z} U}{e^{2k}} + \frac{e^{2U} \log(e)^2 \frac{\partial}{\partial z} U^2}{2e^{2k}} + \frac{e^{2U} \log(e) \frac{\partial}{\partial \rho} k}{2e^{2k} \rho} - \frac{i e^{2U} \log(e) \frac{\partial}{\partial z} k}{2e^{2k} \rho} + \frac{e^{6U} \frac{\partial}{\partial \rho} A^2}{8e^{2k} \rho^2} - \frac{i e^{6U} \frac{\partial}{\partial \rho} A \frac{\partial}{\partial z} A}{4e^{2k} \rho^2} - \frac{e^{6U} \frac{\partial}{\partial z} A^2}{8e^{2k} \rho^2}$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{e^{4U} \log(e) \frac{\partial}{\partial \rho} A \frac{\partial}{\partial \rho} U}{e^{2k} \rho} + \frac{e^{2U} \log(e) \frac{\partial^2}{(\partial \rho)^2} U}{2e^{2k}} + \frac{e^{4U} \log(e) \frac{\partial}{\partial z} A \frac{\partial}{\partial z} U}{e^{2k} \rho} + \frac{e^{2U} \log(e) \frac{\partial^2}{(\partial z)^2} U}{2e^{2k}} + \frac{e^{2U} \log(e) \frac{\partial}{\partial \rho} U}{2e^{2k} \rho} + \frac{e^{6U} \frac{\partial}{\partial \rho} A^2}{4e^{2k} \rho^2} + \frac{e^{4U} \frac{\partial^2}{(\partial \rho)^2} A}{4e^{2k} \rho} + \frac{e^{6U} \frac{\partial}{\partial z} A^2}{4e^{2k} \rho^2} + \frac{e^{4U} \frac{\partial^2}{(\partial z)^2} A}{4e^{2k} \rho} - \frac{e^{4U} \frac{\partial}{\partial \rho} A}{4e^{2k} \rho^2}$$

$$\Lambda = -\frac{e^{2U} \log(e)^2 \frac{\partial}{\partial \rho} U^2}{12e^{2k}} - \frac{e^{2U} \log(e)^2 \frac{\partial}{\partial z} U^2}{12e^{2k}} + \frac{e^{2U} \log(e) \frac{\partial^2}{(\partial \rho)^2} U}{12e^{2k}} + \frac{e^{2U} \log(e) \frac{\partial^2}{(\partial z)^2} U}{12e^{2k}} - \frac{e^{2U} \log(e) \frac{\partial^2}{(\partial \rho)^2} k}{12e^{2k}} - \frac{e^{2U} \log(e) \frac{\partial^2}{(\partial z)^2} k}{12e^{2k}} + \frac{e^{2U} \log(e) \frac{\partial}{\partial \rho} U}{12e^{2k} \rho} + \frac{e^{6U} \frac{\partial}{\partial \rho} A^2}{48e^{2k} \rho^2} + \frac{e^{6U} \frac{\partial}{\partial z} A^2}{48e^{2k} \rho^2}$$

Weyl Tensor Components

$$\begin{aligned} \Psi_0 = & \frac{e^{2U} \log(e)^2 \frac{\partial}{\partial \rho} U^2}{e^{2k}} + \frac{2i e^{2U} \log(e)^2 \frac{\partial}{\partial \rho} U \frac{\partial}{\partial z} U}{e^{2k}} - \frac{e^{2U} \log(e)^2 \frac{\partial}{\partial z} U^2}{e^{2k}} - \frac{e^{2U} \log(e)^2 \frac{\partial}{\partial \rho} U \frac{\partial}{\partial \rho} k}{e^{2k}} - \frac{i e^{2U} \log(e)^2 \frac{\partial}{\partial z} U \frac{\partial}{\partial \rho} k}{e^{2k}} - \frac{i e^{2U} \log(e)^2 \frac{\partial}{\partial \rho} U \frac{\partial}{\partial z} k}{e^{2k}} \\ & + \frac{e^{2U} \log(e)^2 \frac{\partial}{\partial z} U \frac{\partial}{\partial z} k}{e^{2k}} - \frac{3e^{4U} \log(e) \frac{\partial}{\partial \rho} A \frac{\partial}{\partial \rho} U}{2e^{2k} \rho} - \frac{3i e^{4U} \log(e) \frac{\partial}{\partial z} A \frac{\partial}{\partial \rho} U}{2e^{2k} \rho} + \frac{e^{2U} \log(e) \frac{\partial^2}{(\partial \rho)^2} U}{2e^{2k}} + \frac{i e^{2U} \log(e) \frac{\partial^2}{\partial \rho \partial z} U}{e^{2k}} - \frac{3i e^{4U} \log(e) \frac{\partial}{\partial \rho} A \frac{\partial}{\partial z} U}{2e^{2k} \rho} \\ & + \frac{3e^{4U} \log(e) \frac{\partial}{\partial z} A \frac{\partial}{\partial z} U}{2e^{2k} \rho} - \frac{e^{2U} \log(e) \frac{\partial^2}{(\partial z)^2} U}{2e^{2k}} + \frac{e^{4U} \log(e) \frac{\partial}{\partial \rho} A \frac{\partial}{\partial \rho} k}{2e^{2k} \rho} + \frac{i e^{4U} \log(e) \frac{\partial}{\partial z} A \frac{\partial}{\partial \rho} k}{2e^{2k} \rho} + \frac{i e^{4U} \log(e) \frac{\partial}{\partial \rho} A \frac{\partial}{\partial z} k}{2e^{2k} \rho} - \frac{e^{4U} \log(e) \frac{\partial}{\partial z} A \frac{\partial}{\partial z} k}{2e^{2k} \rho} \\ & + \frac{e^{2U} \log(e) \frac{\partial}{\partial \rho} k}{2e^{2k} \rho} + \frac{i e^{2U} \log(e) \frac{\partial}{\partial z} k}{2e^{2k} \rho} + \frac{e^{6U} \frac{\partial}{\partial \rho} A^2}{4e^{2k} \rho^2} - \frac{e^{4U} \frac{\partial^2}{(\partial \rho)^2} A}{4e^{2k} \rho} - \frac{i e^{4U} \frac{\partial^2}{\partial \rho \partial z} A}{2e^{2k} \rho} + \frac{i e^{6U} \frac{\partial}{\partial \rho} A \frac{\partial}{\partial z} A}{2e^{2k} \rho^2} - \frac{e^{6U} \frac{\partial}{\partial z} A^2}{4e^{2k} \rho^2} + \frac{e^{4U} \frac{\partial^2}{(\partial z)^2} A}{4e^{2k} \rho} + \frac{e^{4U} \frac{\partial}{\partial \rho} A}{4e^{2k} \rho^2} + \frac{i e^{4U} \frac{\partial}{\partial z} A}{4e^{2k} \rho^2} \end{aligned}$$

$$\Psi_1 = 0$$

$$\begin{aligned}\Psi_2 = & -\frac{e^{2U} \log(e)^2 \frac{\partial}{\partial \rho} U^2}{3 e^{2k}} - \frac{e^{2U} \log(e)^2 \frac{\partial}{\partial z} U^2}{3 e^{2k}} + \frac{i e^{4U} \log(e) \frac{\partial}{\partial z} A \frac{\partial}{\partial \rho} U}{2 e^{2k} \rho} - \frac{e^{2U} \log(e) \frac{\partial^2}{(\partial \rho)^2} U}{6 e^{2k}} - \frac{i e^{4U} \log(e) \frac{\partial}{\partial \rho} A \frac{\partial}{\partial z} U}{2 e^{2k} \rho} \\ & - \frac{e^{2U} \log(e) \frac{\partial^2}{(\partial z)^2} U}{6 e^{2k}} + \frac{e^{2U} \log(e) \frac{\partial^2}{(\partial \rho)^2} k}{6 e^{2k}} + \frac{e^{2U} \log(e) \frac{\partial^2}{(\partial z)^2} k}{6 e^{2k}} + \frac{e^{2U} \log(e) \frac{\partial}{\partial \rho} U}{3 e^{2k} \rho} + \frac{e^{6U} \frac{\partial}{\partial \rho} A^2}{12 e^{2k} \rho^2} + \frac{e^{6U} \frac{\partial}{\partial z} A^2}{12 e^{2k} \rho^2} - \frac{i e^{4U} \frac{\partial}{\partial z} A}{4 e^{2k} \rho^2}\end{aligned}$$

$$\Psi_3 = 0$$

$$\begin{aligned}\Psi_4 = & \frac{e^{2U} \log(e)^2 \frac{\partial}{\partial \rho} U^2}{e^{2k}} - \frac{2i e^{2U} \log(e)^2 \frac{\partial}{\partial \rho} U \frac{\partial}{\partial z} U}{e^{2k}} - \frac{e^{2U} \log(e)^2 \frac{\partial}{\partial z} U^2}{e^{2k}} - \frac{e^{2U} \log(e)^2 \frac{\partial}{\partial \rho} U \frac{\partial}{\partial \rho} k}{e^{2k}} + \frac{i e^{2U} \log(e)^2 \frac{\partial}{\partial z} U \frac{\partial}{\partial \rho} k}{e^{2k}} + \frac{i e^{2U} \log(e)^2 \frac{\partial}{\partial \rho} U \frac{\partial}{\partial z} k}{e^{2k}} \\ & + \frac{e^{2U} \log(e)^2 \frac{\partial}{\partial z} U \frac{\partial}{\partial z} k}{e^{2k}} + \frac{3 e^{4U} \log(e) \frac{\partial}{\partial \rho} A \frac{\partial}{\partial \rho} U}{2 e^{2k} \rho} - \frac{3i e^{4U} \log(e) \frac{\partial}{\partial z} A \frac{\partial}{\partial \rho} U}{2 e^{2k} \rho} + \frac{e^{2U} \log(e) \frac{\partial^2}{(\partial \rho)^2} U}{2 e^{2k}} - \frac{i e^{2U} \log(e) \frac{\partial^2}{\partial \rho \partial z} U}{e^{2k}} - \frac{3i e^{4U} \log(e) \frac{\partial}{\partial \rho} A \frac{\partial}{\partial z} U}{2 e^{2k} \rho} \\ & - \frac{3 e^{4U} \log(e) \frac{\partial}{\partial z} A \frac{\partial}{\partial z} U}{2 e^{2k} \rho} - \frac{e^{2U} \log(e) \frac{\partial^2}{(\partial z)^2} U}{2 e^{2k}} - \frac{e^{4U} \log(e) \frac{\partial}{\partial \rho} A \frac{\partial}{\partial \rho} k}{2 e^{2k} \rho} + \frac{i e^{4U} \log(e) \frac{\partial}{\partial z} A \frac{\partial}{\partial \rho} k}{2 e^{2k} \rho} + \frac{i e^{4U} \log(e) \frac{\partial}{\partial \rho} A \frac{\partial}{\partial z} k}{2 e^{2k} \rho} + \frac{e^{4U} \log(e) \frac{\partial}{\partial z} A \frac{\partial}{\partial z} k}{2 e^{2k} \rho} \\ & + \frac{e^{2U} \log(e) \frac{\partial}{\partial \rho} k}{2 e^{2k} \rho} - \frac{i e^{2U} \log(e) \frac{\partial}{\partial z} k}{2 e^{2k} \rho} + \frac{e^{6U} \frac{\partial}{\partial \rho} A^2}{4 e^{2k} \rho^2} + \frac{e^{4U} \frac{\partial^2}{(\partial \rho)^2} A}{4 e^{2k} \rho} - \frac{i e^{4U} \frac{\partial^2}{\partial \rho \partial z} A}{2 e^{2k} \rho} - \frac{i e^{6U} \frac{\partial}{\partial \rho} A \frac{\partial}{\partial z} A}{2 e^{2k} \rho^2} - \frac{e^{6U} \frac{\partial}{\partial z} A^2}{4 e^{2k} \rho^2} - \frac{e^{4U} \frac{\partial^2}{(\partial z)^2} A}{4 e^{2k} \rho} - \frac{e^{4U} \frac{\partial}{\partial \rho} A}{4 e^{2k} \rho^2} + \frac{i e^{4U} \frac{\partial}{\partial z} A}{4 e^{2k} \rho^2}\end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "Unknown"